

Scaling Book Exercises: Section 12, Quiz 2

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Quiz: GPU Nodes

Source scan: ../handwritten/section_12_handwritten.pdf, pages 3–4

Reference: <https://jax-ml.github.io/scaling-book/gpus/>

Question 1: Total bandwidth of an H100 node

Four NVSwitches with 64 ports each and 25 GB/s per port provide

$$4 \cdot 64 \cdot 25 = \boxed{6.4 \text{ TB/s}}$$

of switch capacity. The eight GPUs can each egress only 450 GB/s:

$$8 \cdot 450 = \boxed{3.6 \text{ TB/s}}.$$

The smaller value is the bottleneck, so the usable aggregate node bandwidth is

$$\boxed{3.6 \text{ TB/s}}.$$

Question 2: Bisection bandwidth

Any even bisection has four GPUs on each side. Counting traffic in both directions, all eight GPUs can egress across the cut:

$$\boxed{W_{\text{bisect}} = 8(450 \text{ GB/s}) = 3.6 \text{ TB/s}}.$$

Question 3: Eight-GPU AllGather

Let an array have total size B bytes and let $N = 8$. Each GPU begins with B/N bytes and sends one such chunk in each of $N - 1$ ring steps. With per-GPU egress bandwidth $W = 450 \text{ GB/s}$,

$$T_{\text{AG}} = \frac{(N - 1)B}{NW} = \frac{7B}{8(450 \times 10^9)} \approx \frac{B}{450 \times 10^9}.$$

For `bf16`[D_X, F] with $D = 4096$ and $F = 65,536$,

$$B = 2DF = 2(4096)(65,536) = 536,870,912 \text{ bytes}.$$

Therefore

$$\boxed{T_{\text{AG}} = \frac{7(536,870,912)}{8(450 \times 10^9)} \approx 1.04 \text{ ms}}.$$

The large- N approximation gives 1.19 ms; latency and imperfect bandwidth utilization can raise the observed time to roughly 1.5 ms.